Progress Report: Bangla–English Multimodal Fake News Detection — XLM-RoBERTa Text Classifier on Fakeddit

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Abstract—This report documents progress on a Bangla–English text-based fake news classifier built for the TruthLens AR project. Using the Fakeddit dataset (17,045 labeled Reddit posts), an XLM-RoBERTa (xlm-roberta-base) sequence classifier was fine-tuned via the Hugging Face Trainer API on Google Colab. The model reached 83.1% validation accuracy and an F1-score of 0.788 after three epochs, and was saved as a reusable checkpoint. A separate step-by-step walkthrough (tokenization, embeddings, 12-layer encoder, classification head) was built to inspect and explain individual predictions, directly supporting the explainability goal of the TruthLens architecture.

I. Objective

The goal of this stream of work is to produce a lightweight, multilingual text classifier that can flag likely fake news headlines/titles as a standalone signal within TruthLens, and to build an interpretable inference pipeline that exposes intermediate representations (tokens, embeddings, hidden states, logits) for later explainability features. XLM-RoBERTa was chosen for its native multilingual, including Bangla, subword vocabulary, avoiding the need for a separate Bangla-only encoder.

II. System Configuration

Google Colab with GPU runtime; Python 3 environment with transformers, datasets, torch, scikit-learn, pandas, and accelerate installed via pip. Data (final_train_17k.tsv and an associated images.zip, unused in this stream) mounted from Google Drive at /content/drive/MyDrive/499A_FakeNews.

III. Work Completed

A. Data Loading and Cleaning

The Fakeddit-derived TSV (17,045 rows, 16 columns, including author, domain, image metadata, and three label granularities) was loaded with pandas. Only the clean_title text field and the binary 2_way_label were retained and renamed to text/label, giving a mildly imbalanced set of 10,404 real- and 6,641 fake-labeled samples.

B. Train/Test Split and Tokenization

An 80/20 stratified split (random_state=42) produced 13,636 training and 3,409 test examples. Both splits were converted to Hugging Face Dataset objects and tokenized with the xlm-roberta-base tokenizer, padding to max_length with truncation at 128 tokens.

C. Model Fine-Tuning

AutoModelForSequenceClassification was initialized from xlm-roberta-base with a fresh two-class classification head; the base model's pretrained pooler/LM-head weights were discarded, as expected for this task change. Training used the Trainer API with per-device batch size 16, three epochs, epoch-level evaluation and checkpointing, and load_best_model_at_end enabled; accuracy, F1, precision, and recall were computed each epoch via a custom compute_metrics function.

D. Model Saving

The fine-tuned model and tokenizer were exported to Google Drive (499A_FakeNews/text_model) as config.json, model.safetensors, tokenizer_config.json, and tokenizer.json, making the checkpoint reloadable independently of the training notebook.

E. Interpretability Walkthrough

A separate inspection script runs a sample headline through five explicit stages: subword tokenization, token-to-ID mapping, raw embedding lookup (768-d), full 12-layer encoder pass, and finally the saved classifier head, printing the tensor at every stage. On the test sentence "Scientists discover new planet capable of supporting life" the pipeline produced a 77.7% fake-probability, illustrating both the model's current bias toward flagging sensational science headlines and the value of exposing intermediate states for future explainability features in TruthLens.

IV. Results

Table 1 summarizes epoch-wise validation performance. Accuracy and F1 improved steadily over three epochs, while validation loss began to plateau between epochs 2 and 3, suggesting the model is close to converging for this text-only setup and that further gains will likely require more epochs, learning-rate tuning, or additional data/modalities rather than simply longer training on the current configuration.

Table 1: XLM-RoBERTa Fine-Tuning — Epoch-Wise Metrics

Ep.	Tr.Loss	Val.Loss	Acc.	F1	Prec.	Rec.
1	0.567	0.548	0.758	0.745	0.633	0.904
2	0.438	0.406	0.818	0.777	0.744	0.814
3	0.343	0.411	0.831	0.788	0.769	0.809

V. Problems Identified

The classifier currently uses only the clean_title text field, ignoring the image and richer metadata columns present in the source TSV (domain, hasImage, image_url, score, upvote_ratio), so it cannot yet exploit visual or engagement-based fake-news signals that the wider TruthLens architecture is meant to fuse in. The fake-probability example above also hints at a possible bias toward sensational/scientific phrasing that will need broader qualitative testing before the checkpoint is trusted for demo use.

VI. Current Status

Data preparation, tokenization, fine-tuning, checkpoint saving, and the interpretability walkthrough are all complete for the text-only XLM-RoBERTa model, which now stands as a validated, ready-to-integrate text signal for TruthLens. Fusion with image and metadata modalities, and integration into the TruthLens VLA + AR pipeline, are not yet started.

VII. Planned Work

Next steps are to (1) evaluate the saved checkpoint on additional, more diverse Bangla and English headlines to probe for topic-specific bias, (2) add confusion-matrix and per-class error analysis beyond the aggregate accuracy/F1/precision/recall already logged, (3) wrap the five-stage interpretability walkthrough into a reusable function that can surface attention/embedding diagnostics inside the TruthLens explainability layer, and (4) begin fusing this text signal with the image-based and multimodal components already under development by the rest of the team.

VIII. Conclusion

This stream has produced a working, reasonably accurate (83.1%) XLM-RoBERTa checkpoint with an interpretable, stage-by-stage inference pipeline, giving TruthLens a validated multilingual text signal. The priority for the next phase is broader evaluation and fusing this text model with the image/metadata modalities under development by the rest of the team.